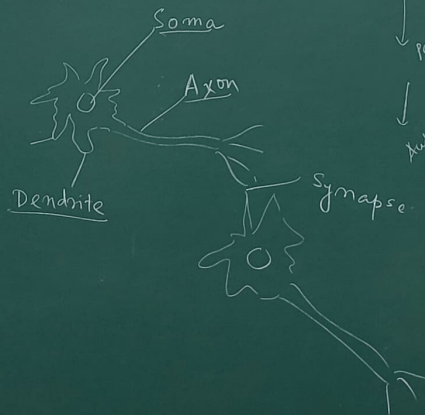


Neural Networks

- 1873, 1890 $\Rightarrow$ Alexander Bain & William James
- 1943 $\Rightarrow$ Warren McCulloch & Walter Pitts
- 1957 $\Rightarrow$ Frank Rosenblatt
- 1969 $\Rightarrow$ Minsky & Papert

Featurization

$X \Rightarrow$ 



Low Data, $X = \{x_1, x_2, \dots, x_m\}$

2013-2014

↓ manual
Feature
Extraction

↓ PCA, LDA
B...

↓ Automatic FE

AAAT

Neural Networks

- 1873, 1890 $\Rightarrow$ Alexander Bain & William James
Conceptual

Neuroscientist

- 1943 $\Rightarrow$ Warren McCulloch & Walter Pitts
Theoretical Foundation of Biological Nervous System Logician

- 1957 $\Rightarrow$ Frank Rosenblatt
Perceptron, + MCP

- 1969 $\Rightarrow$ Minsky & Papert

XOR

The Perceptron

AND, OR, XOR

$$X = \{x_1, x_2, \dots, x_N\}$$



Signum:
$$\text{Sgn}(z) = \begin{cases} -1, & z < 0 \\ 0, & z = 0 \\ 1, & z > 0 \end{cases}$$

$$\text{Sgn}(z) = \begin{cases} 1, & z \geq 0 \\ 0, & z < 0 \end{cases}$$

$$x_2 = -x_1 + 0.5$$

OR $\text{Sgn}(x_1 + x_2 - 0.5)$

	x_1	x_2
0	0	0
0	0	1
1	1	0
1	1	1

0	-0.5	0
1	0.5	1
1	0.5	1
1	1.5	1

AAAA

Neural Networks

- 1873, 1890 $\Rightarrow$ Alexander Bain & James
Conceptual

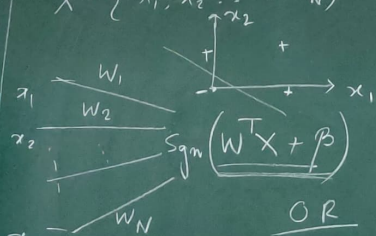
- 1943 $\Rightarrow$ Warren McCulloch & Walter Pitts
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Perceptron

- 1969
Minsky & Papert

The Perceptron

$$X = \{x_1, x_2, \dots, x_N\}$$



OR

x_1	x_2
0	0
0	1
1	0
1	1

AND, OR, XOR

Signum:
$$\text{Sgn}(z) = \begin{cases} -1, & z < 0 \\ 0, & z = 0 \\ 1, & z > 0 \end{cases}$$

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$$x_2 = -x_1 + 0.5$$

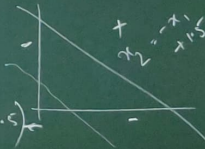
OR $\text{Sgn}(x_1 + x_2 - 0.5)$

-0.5	0
0.5	1
0.5	1
1.5	1

$\text{Sgn}(x_1 + x_2 - 1.5)$

0
0
0
1

x_1	x_2	$x_1 \cdot x_2$
0	0	0
0	1	0
1	0	0
1	1	1



AAAI

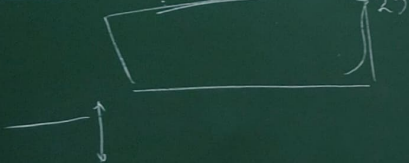
Neural Networks

- 1873, 1890 $\Rightarrow$ Alexander Bain & William James
Conceptual.

- 1943 $\Rightarrow$ Warren McCulloch & Walter Pitts
Theoretical Foundation of Biological Nervous System Logic
Neuroscientist

- 1957 $\Rightarrow$ Frank Rosenblatt $\checkmark$
Perceptron, + MCP

- 1969 $\Rightarrow$ Minsky & Papert

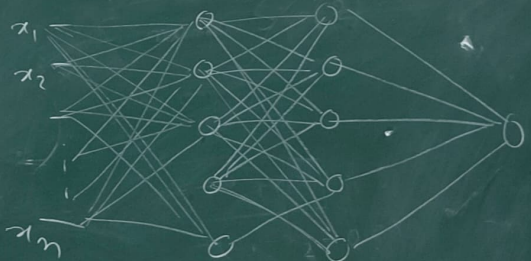


XOR

x_1	x_2	$x_1 \oplus x_2$
0	0	0
0	1	1
1	0	1
1	1	0



Activation Function / Transfer Function
Depth, Width



AAAT

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XOR

x_1	x_2
0	0
0	1
1	0
1	1

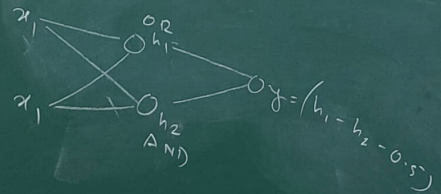
hidden state
$h_1 = \text{Sgn}(x_1 + x_2 - 0.5)$
0
1
1
1

Activation Function / Transfer Function

Depth, Width
$h_2 = \text{Sgn}(x_1 + x_2 - 1.5)$
0
0
0
1

Transfer Function

Output
$y = \text{Sgn}(h_1 - h_2 - 0.5)$
0
1
1
0



AAAT

Neural Networks

Feature-Extractor



Input → ~~Fixed Feature Extractor~~

XOR

hidden state
 $h_1 = \text{Sgn}(x_1 + x_2 - 0.5)$

0

~~hard programmed feature~~

Activation Function

$h_2 = \text{Sgn}(x_1 + x_2 - 1.5)$

0

0

0

1

learned Classifier

Depth, Width

Activation function

Transfer Function

output
 $y = \text{Sgn}(h_1 - h_2 - 0.5)$

0

1

1

0

x_1, x_2

AAAT

Neural Networks

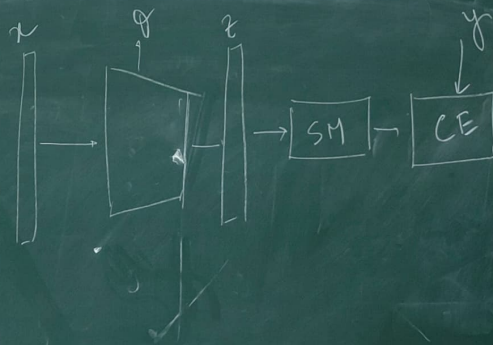
Logistic Regression

Given $X \in \mathbb{R}^d$ $f(x) = \theta^T X$; θ is a $d \times K$ matrix

$C = \{0, 1, \dots, K-1\}_K$

$P_c(x) = \text{Softmax}(f_\theta(x))_c$

$$\text{loss: } -\log p_\theta(y|x) = \frac{\exp(f_\theta(x)_c)}{\sum_{c'} \exp(f_\theta(x)_{c'})}$$



AAAI

Neural Networks

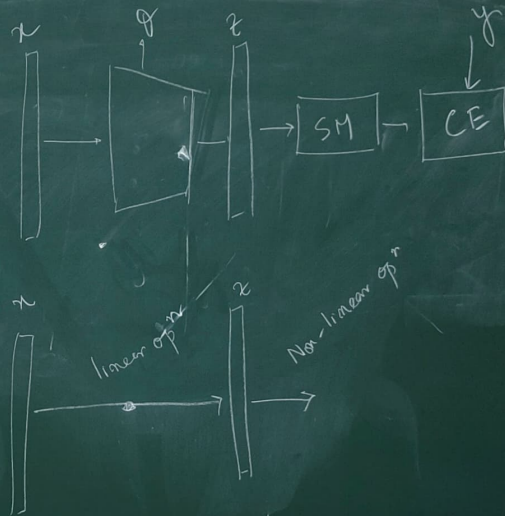
Logistic Regression

Given $X \in \mathbb{R}^d$ $f(x) = \theta^T X$; θ is a $d \times K$ matrix

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$$p_c(x) = \text{Softmax}(f_{\theta}(x))_c$$

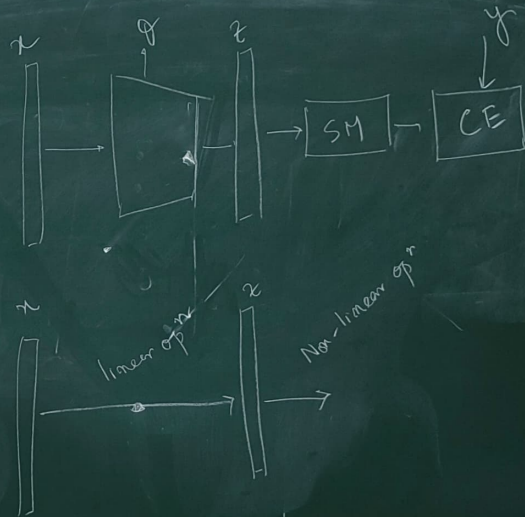
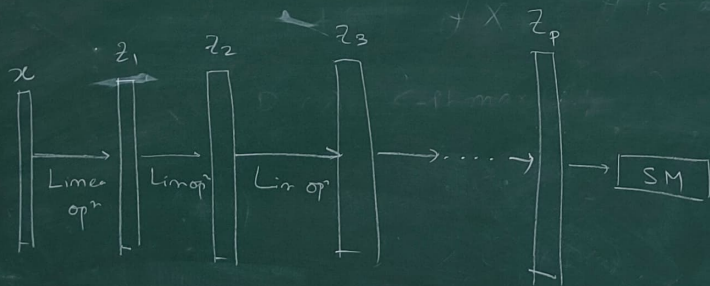
$$\text{loss: } -\log p_{\theta}(y|x) = \frac{\exp(f_{\theta}(x)_c)}{\sum_{c'} \exp(f_{\theta}(x)_{c'})}$$



AAAT

Neural Networks

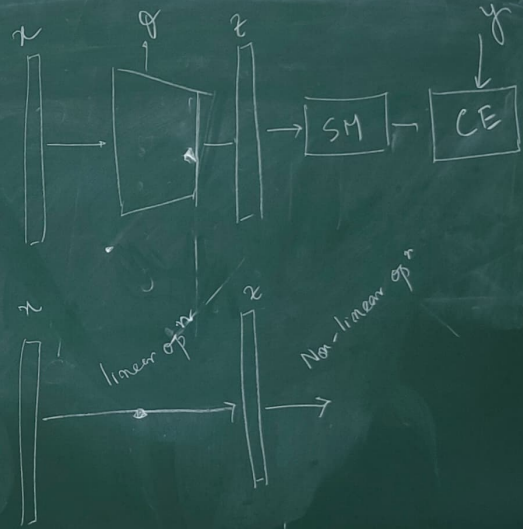
NN-LR-Attempt 1



AAAT

Neural Networks

NN-LP-Attempt 2



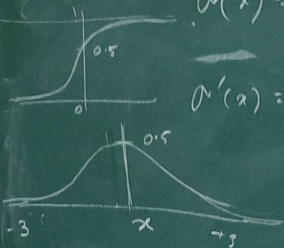
AAAT

Neural Networks

① Sigmoid

$$a(x) = \frac{1}{1 + e^{-x}}$$

$$a'(x) = a(x)(1 - a(x))$$

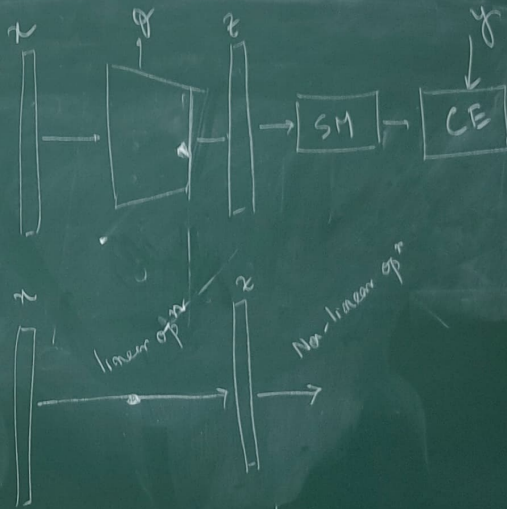
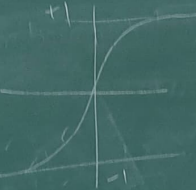


if $x \geq 3$
 $x \leq -3$ } $a'(x) \rightarrow 0$

② $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

$$f'(x) = 1 - \tanh^2(x)$$

Not symmetric around zero



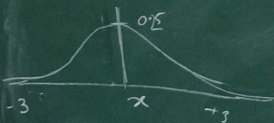
AAAT

Neural Networks

① Sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$



$$\left. \begin{array}{l} \text{if } x \geq 3 \\ x \leq -3 \end{array} \right\} \sigma'(x) \rightarrow 0$$

② tanh(x)

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$f'(x) = 1 - \tanh^2(x)$$

Not symmetric around zero

③ Softmax

$$\frac{\exp(x)}{\sum \exp(x)}$$



④ ReLU: Rectified Linear Unit

$$= \max(0, x)$$

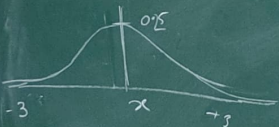
AAAT

Neural Networks

① Sigmoid

$$a(x) = \frac{1}{1 + e^{-x}}$$

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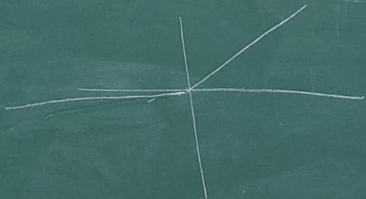
$$\left. \begin{array}{l} \text{if } x \geq 3 \\ x \leq -3 \end{array} \right\} a'(x) \rightarrow 0$$

② tanh(x)

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$f'(x) = 1 - \tanh^2(x)$$

Not symmetric around zero



③ Leaky ReLU

$$f(x) = \begin{cases} x & x \geq 0 \\ 0.01x & x \leq 0 \end{cases}$$

④ Softmax

$$\frac{\exp(x)}{\sum \exp(x)}$$



⑤ ReLU: Rectified Linear Unit
= max(0, x)

1

0

ReLU

0

0

0

0

0

0

0

0

0

0

0

0

0

0

0

AAAT

Neural Networks

① Sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$



$$\text{if } \begin{cases} x \geq 3 \\ x \leq -3 \end{cases} \left\{ \begin{array}{l} \sigma'(x) \rightarrow 0 \end{array} \right.$$

② tanh(x) = $\frac{e^x - e^{-x}}{e^x + e^{-x}}$

③ ELU

Exponential Linear Unit

$$f(x) = \begin{cases} x & x > 0 \\ a(e^x - 1) & \text{otherwise} \end{cases}$$

learnable parameter

④ Concatenated ReLU
 $= [\text{ReLU}(x) \oplus \text{ReLU}(-x)]$

⑤ Softmax

$$\frac{\exp(x_i)}{\sum_j \exp(x_j)}$$



⑥ ReLU: Rectified Linear Unit

⑦ Parameter ReLU

$$f(x) = \begin{cases} x & x > 0 \\ a \cdot x & \text{otherwise} \end{cases}$$

a is a learnable parameter

AAAT

Neural Networks

GELU : Gaussian Error Linear Unit

$$= x \cdot \Phi(x)$$

$$= x \cdot P(X \leq x)$$

Swiss GLU

$$= x \cdot \sigma(\beta x)$$

GLU : Gated Linear Unit

$$x \odot \sigma(wx + b)$$

elementwise product

Hard tanh

$$\begin{cases} -1 & x < -1 \\ 1 & x > 1 \\ x & \text{other} \end{cases}$$

